

General Exam January 9, 2008

Justify all your answers fully with a complete explanation on a separate sheet of paper for each problem, but put numerical answers in the boxes on the exam sheet. Keep the problems in correct order when you turn in your answers.

Problem 1

- (a) Show that a group G will have outer automorphisms (automorphisms which are not inner) if it can be properly imbedded as a normal subgroup $G \triangleleft G'$ of a group in such a way that $G \cdot \text{Centralizer}_{G'}(G) \neq G'$.
- (b) Show that A_n has outer automorphisms whenever $n \geq 4$.
- (c) Explain the mantra "Every automorphism of G is inner, somewhere."

Problem 2

- (a) Let R be a commutative ring with 1, and M a finite direct sum $M = M_1 \oplus \cdots \oplus M_n$ of simple unital left R -modules M_i . Show that M has both the ascending and descending chain conditions on R -submodules. (b) Where does your argument use the hypotheses that R is unital, R is commutative, or M is unital?

Problem 3

Let V be an n -dimensional vector space V over a finite field F of q elements.

- (a) Show that the number of invertible linear operators on V is $\prod_{i=0}^{n-1} (q^n - q^i)$.
- (b) Find the cardinality $|P|$ of any 3-Sylow subgroup $P \leq G = \text{GL}(4, \mathbb{F}_{81})$ where $n = 4$ and $F = \mathbb{F}_{81}$ is the field of $q = 81$ elements.

$$|P| = \square$$

- (c) Find the cardinality $|P'|$ of any 3-Sylow subgroup $P' \leq G'$ of the special linear group $G' = \text{SL}(4, \mathbb{F}_{81})$ (those invertible 4×4 matrices of determinant 1) over a field $\mathbb{F}_{81}$ elements.

$$|P'| = \square$$

(d) Describe up to isomorphism (in terms of Jordan canonical forms) all possible 3-torsion elements T of the general linear group $\text{GL}(4, \mathbb{F}_{81})$: all invertible operators T with $T^{3^e} = Id$ for some $e \geq 0$. For each T list its minimum polynomial $\mu_T(x)$ and its 3-period (the smallest $e \geq 0$ with $T^{3^e} = Id$). [Hint: all eigenvalues of T already lie in $\mathbb{F}_3$.]

Problem 4

(a) If $r \in \mathbb{Q}$ is a rational root of a monic integral polynomial

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$$

show that $r \in \mathbb{Z}$ is integral.

(b) Factor $y^5x + y^3x^2 + y + x^3$ into irreducible factors in $\mathbb{Z}[x, y]$, explaining why each is irreducible.

Problem 5

Let R be a unital commutative ring, and consider 5 possible properties such a ring might have: it is $(\mathcal{P}_1)$ a domain, $(\mathcal{P}_2)$ a PID, $(\mathcal{P}_3)$ Euclidean, $(\mathcal{P}_4)$ noetherian, $(\mathcal{P}_5)$ a UFD.

(a) For which n is it true that $R[x]$ always inherits property $(\mathcal{P}_n)$ from R (if R has $(\mathcal{P}_n)$, so must $R[x]$)? Circle the n 's for which this holds, and explain your answer or give a counterexample.

$$n = \begin{matrix} 1 & 2 & 3 & 4 & 5 \\ \hline \end{matrix}$$

(c) For which n is it true that R inherits property $(\mathcal{P}_n)$ from $R[x]$? Circle the n 's for which this holds, and explain your answer or give a counterexample.

$$n = \begin{matrix} 1 & 2 & 3 & 4 & 5 \\ \hline \end{matrix}$$

(d) For which $n \neq 5$ is it true that a quotient R/P of R by a proper prime ideal ($P \triangleleft R, P \neq R$) inherits property $(\mathcal{P}_n)$ from R ? Circle the n 's for which this holds, and explain your answer or give a counterexample.

$$n = \begin{matrix} 1 & 2 & 3 & 4 \\ \hline \end{matrix}$$

Problem 6

Let $R = \mathbb{Z} + 2\mathbb{Z}[x] = \mathbb{Z}1 + \sum_{n=0}^{\infty} 2\mathbb{Z}x^n$. (a) Show that R is not a UFD by finding an irreducible element that is not prime. (b) Show that R is not noetherian by showing that the ideal $2\mathbb{Z}[x]$ is not finitely generated: $2\mathbb{Z}[x] \neq \sum_{i=1}^n Rf_i(x)$ for any $f_i(x) \in 2\mathbb{Z}[x]$.

Problem 7

Let F be a field.

- (a) Show that if $a \in E$ is an element of a field extension of F with $[F(a) : F] = 7$, then $F(a^3) = F(a)$.
(b) Show that any subgroup of order 8 of the multiplicative group $F^\times$ of the field F must be cyclic. Is this also true of subgroups of order 16 ?

Problem 8

- (a) Find a splitting field E of the polynomial $x^4 + 3x^3 + 4x^2 + 3x + 3$ over the rationals $F = \mathbb{Q}$, and find its degree $[E : F]$. [Hint: write $E = F(\alpha, \beta)$ for an easy pure imaginary α and a real β .]
(b) Find the Galois group $\text{Gal}(E/F)$ of the extension field (describe all the automorphisms by their actions on α, β).
(c) Diagram the lattice of subgroups of the Galois group and the corresponding lattice of sub-field-extensions of E/F .